MSCI223: Business Modelling and Simulation
2023/24 Group Coursework Task

Task instructions (please read carefully)

- The coursework consists of a business case that requires the use of simulation modelling.
- There are 5 questions associated with this case, covering the range of a simulation project.
- The maximum score is 100.
- You are expected to use appropriate software (e.g. Witness, Stat::Fit) to help you answer the questions.
- This coursework is to be done as part of a group of three or four.
- Each group has a unique dataset, which can be found on Moodle. Each group is labelled using a letter. Use the dataset corresponding to this letter.
- Teamwork! I suggest you do NOT divide questions among group members as these questions will use the same base simulation model. You all need to participate in building the simulation. In your submission ZIP-file, please also include a one-page statement on each member's contribution in the coursework.
- Collusion between groups and plagiarism are regarded as serious offences and will be penalised severely. Information about the university's plagiarism framework can be found here: https://modules.lancaster.ac.uk/mod/resource/view.php?id=1384800.

Submission instructions

- The deadline for submission is **4:00PM on May 3rd 2024**.
- Only **one** member of the group is required to make the submission.
- Your answers to the questions must be typed (e.g. using Microsoft Word). The maximum length of this document should be **15 pages**.
- **You should include full details of your methods in your answers**. For each question, if you have used a new model or a new experiment, submit the corresponding .mod or .wexp file, named appropriately. Please also include enough explanation in your write up to make it clear what experimentation has been performed and how these results were analysed.
- Place all files in a .zip folder to submit. Please do not use any other compression software (such as RAR or ARJ).

Congress Office Chair Plant – System Description

Congress is a company that manufactures office furniture. One site is dedicated to a particular office chair. The plant takes a fabric sheet, a set of bars for the frame and a pre-made foam cushion and assembles these into the office chair. The chairs are also stored in a small warehouse onsite (total capacity is 100 chairs). This warehouse also acts as a distribution centre – as orders come in, they are processed before the correct number of chairs are collected ready to be shipped to the purchaser. Operations take place between 9am and 5pm, seven days a week. All staff take a break between 12:30pm and 1pm for lunch. You may assume that any processes still going on at 12:30pm or 5pm are left as they are and picked up again when the shift continues.

Fabric Sheet Delivery and Processing

The fabric sheets are delivered every two days, at 8am, with a delivery of 150 sheets. The capacity of the inventory for fabric sheets is 500. If the delivery of 150 sheets takes the capacity beyond 500, the remaining are returned at no cost.

The fabric sheet first needs to be cut to the correct size and shape. This is done using one of two cutting machines. An upholsterer (a fabric worker within the factory) first loads the sheet into the machine. The cutting process is then automated and takes exactly 7 minutes. The upholsterer is then needed to unload the cut sheet and take it to a set of shelves (which only has space for 10 pieces of fabric). It waits there until an upholsterer is ready to do the sewing work to make the chair cover. This can be at one of three sewing machines. Note that there are currently only three upholsterers working on the site. After this, they are placed in a buffer with space for 50 covers to await the final assembly.

Frame kit Delivery and Processing

A delivery of 100 sets of metal frame bars occurs everyday at 10am. The storage for these sets can hold 500. As with the fabric sheets, if the delivery takes the capacity beyond 500, the remaining are returned at no cost.

The sets must first be welded together. There are three welding stations, and the full process takes on average 15 minutes. After this they are placed in a buffer that can store 15 completed frames, ready for assembly.

Cushion Delivery

The cushions themselves are delivered in the state needed for the chair, so do not undergo any further processing before the full chair is assembled.

The inventory storage space has enough capacity for 600 cushions. They are delivered every 2 days at 3pm, and 150 are delivered. As with the fabric sheets and metal frame bars, if the delivery takes the capacity beyond 600, the remaining are returned at no cost.

Assembly and Warehouse

The assembly is performed by 5 members of staff at 5 assembly stations. It is a long process, requiring the cushions to be secured into the frames and the covers stapled on. The completed chairs are then stored in the warehouse.

The warehouse is split into two sections, one for small orders and one for large orders. Both sections have capacity for 50 chairs. The allocation of a chair to a section is based on which has the most free space.

Orders and Order Processing

There are two types of order: small (which consist of a single chair) and large (which consist of 20 chairs). These can happen at any time of day, as they are made through an online system. Each order must first be processed by a receptionist. They are then added to the list of orders that must be collected together before being shipped. Two separate queues are used here, one for small orders and one for large orders. One worker is assigned to each queue, and it takes a little time to get the chairs from the warehouse, and this can only be done if the chairs are available. Once the full quota of chairs for the order are collected, the order is shipped. At this point, the order is considered complete. The aim of Congress is to complete 95% of orders within 24 hours of it being placed.

Data

The available data include the following:

- 200 chairs were monitored through all the stages of their production. The durations (in minutes) of each process in the manufacturing were monitored and collected;
- The number of large and small orders over a 13-week period;
- For the orders in the last two weeks of that period, the time it took to process the order and the time it took to collect the chairs together were collected (both in minutes).
- A collection of 40 weeks from last year were monitored from a performance perspective. For each week, the average time it took to process orders (small and large separately, both in minutes) and the total number of orders completed was noted.

Overall Task

The company has approached you to model their operations using simulation. They are particularly interested in the delivery policy, as they feel fewer deliveries may be possible to reduce costs. However, there is concern that if this was done, there may be times when the stock runs short, leading to orders not being processed within 24 hours. There are also questions related to whether a quality control procedure should be implemented within the plant.

See the next page for the specific tasks.

<u>Hint</u>

You may find pages 91 and 92 of the Witness reference manual useful for more complex input and output rules.

Question 1: Conceptual Model [25 marks]

Provide a full conceptual model for your simulation model of the plant and its operations, providing all the necessary details. Following this, state any distributions used with a justification for their selection.

Question 2: Validation [15 marks]

Give full details of any validation procedures you used for this model.

Question 3: Inventory Control Deliveries [25 marks]

One of the key questions the company is asking is whether an alternative delivery schedule could be applied for fabric, frame kits and cushions. Each delivery has a fixed charge, regardless of how many items are delivered, so reducing the number of deliveries would reduce the overall cost. The time between deliveries must be a whole number of days and the quantity being delivered must be a multiple of 50.

Come up with alternative plan to improve the costs and evaluate it, providing confidence and prediction intervals for each of the following key performance measures:

a) Mean time between an order being placed and the items being shipped (for both small and large orders); and
b) The percentage of orders shipped within 24 hours of being placed.

Compare your proposed plan with the current plan based on the same measures. Are the differences statistically significant?

Question 4: Analysis of Bottlenecks and Potential Improvements [15 marks]

The company are interested in identifying any bottlenecks that could help the system function more smoothly. Identify one such point (including some evidence for this) and suggest an improvement that could be made. Improvements could be one of the following:

- Adding an additional machine or workstation;
- Adding one additional member of staff;
- Increasing the size of one storage/buffer space;
- Rearranging storage space between large and small orders in the warehouse section or the rules for how chairs are added to each section.

Evaluate your proposal in terms of the performance measures identified above.

Question 5: Quality Control [20 marks]

Congress is concerned about the number of chairs that are being returned due to quality issues – currently about 6% are found to be faulty after shipping. For these, the company bears the cost of delivery and replacing with another chair. One idea is to introduce a Quality Control step into the process, following the assembly but before it goes into the warehouse, with three members of staff being allocated to this. There is uncertainty in exactly how long it would take to do the quality control procedure but the management believe it would be between 3 minutes and 10 minutes, with a most likely value of 5 minutes. Any chairs found to be faulty would be scrapped, rather than going into warehouse.

Modify your simulation model to measure the impact of this on the performance measures seen above. Use the current system settings, i.e. without the changes to the delivery policy or the bottlenecks proposed in Questions 3 and 4.